

# Prediction vs. Inference

## Simple linear regression (SLR)

one predictor ( $x$ ), one response ( $y$ )

$X$ : predictor, independent, covariate, features

$y$ : response, dependent, target

| $X$<br>frog length<br>(cm) | $y$<br>frog wt<br>(g) |
|----------------------------|-----------------------|
| 10                         | 20                    |
| 12                         | 24                    |
| 9                          | 21                    |
| $\vdots$                   | $\vdots$              |

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, i=1, \dots, n$$

$y_i$  =  $i$ th observation of the response  $y$  Random

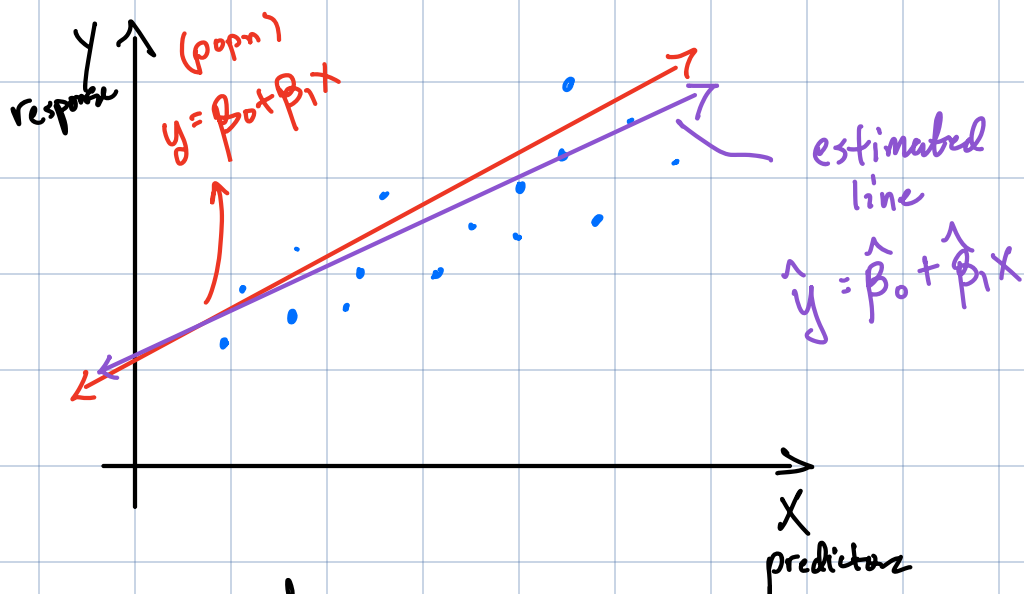
$x_i$  =  $i$ th observation of predictor  $X$  Fixed

$n$  = sample size Fixed

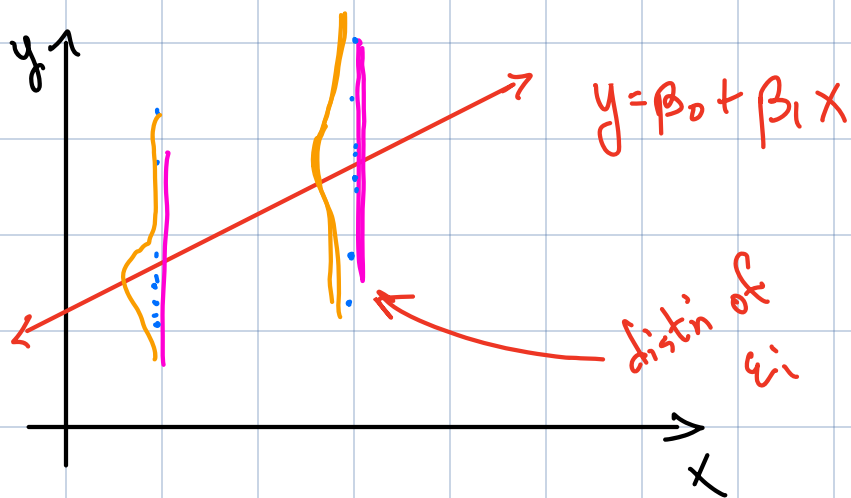
$\beta_0$  = unknown intercept parameter Fixed

$\beta_1$  = unknown slope parameter Fixed

$\varepsilon_i$  = random error term Random



| parameters | estimates       |             |
|------------|-----------------|-------------|
| $\mu$      | $\bar{x}$       | $\hat{\mu}$ |
| $\sigma^2$ | $s^2$           |             |
| $\beta_0$  | $\hat{\beta}_0$ |             |
| $\beta_1$  | $\hat{\beta}_1$ |             |



1. Simple: 1 predictor  $x$
2. linear:  $y_i$  is a linear fctn of  $\beta_0 + \beta_1$

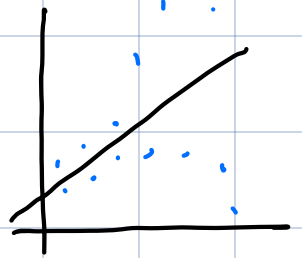
Model Assumptions

1. linearity

2.  $\varepsilon_i$ 's are random variables that satisfy

basic

- (i)  $E(\varepsilon_i) = 0$
- (ii)  $\text{Var}(\varepsilon_i) = \sigma^2$  constant
- (iii)  $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0 \quad i \neq j$   
uncorrelated errors



classical

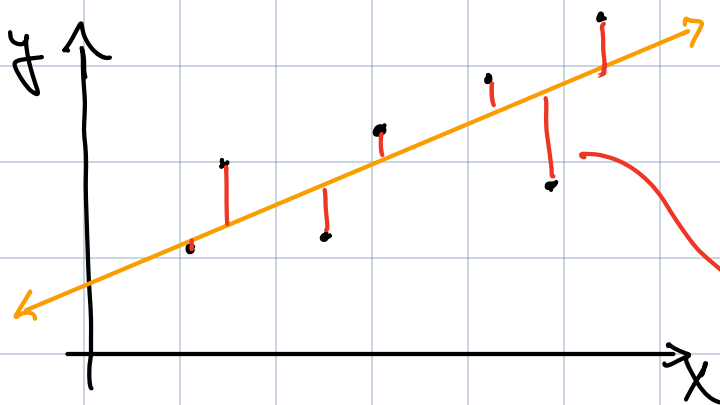
- (iv)  $\varepsilon_i \sim N(0, \sigma^2)$  useful for inference

## Regression Function

$$g(x) = E(y | X=x) = \beta_0 + \beta_1 x$$

Estimate  $\beta_0$  &  $\beta_1$  (SLR)

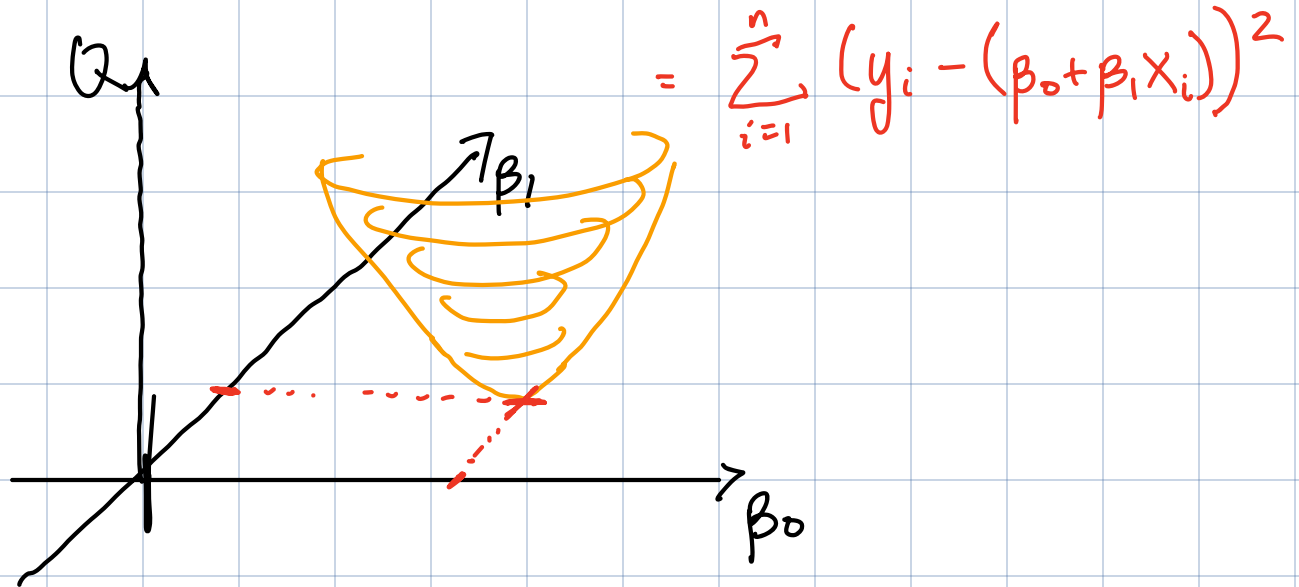
How?



Use least squares  
principle.

Try to minimize  
these in some  
way.

Minimize  $Q(\beta_0, \beta_1) = \sum_{i=1}^n \varepsilon_i^2$



$$\frac{\partial Q}{\partial \beta_0} = 0$$

$$\frac{\partial Q}{\partial \beta_1} = 0$$

$$\frac{\partial Q}{\partial \beta_0} = \frac{\partial}{\partial \beta_0} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

$$= \sum_{i=1}^n \frac{\partial}{\partial \beta_0} (y_i - \beta_0 - \beta_1 x_i)^2$$

$$= \sum_{i=1}^n 2(y_i - \beta_0 - \beta_1 x_i)(0 - 1 - 0)$$

$$= -2 \sum_{i=1}^n y_i - \beta_0 - \beta_1 x_i = 0$$

chain rule

$$\frac{\partial Q}{\partial \beta_1} = \sum_{i=1}^n \frac{\partial}{\partial \beta_1} (y_i - \beta_0 - \beta_1 x_i)^2$$

$$= \sum_{i=1}^n 2(y_i - \beta_0 - \beta_1 x_i)(0 - 0 - x_i)$$

$$= -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) x_i = 0$$

$$\beta_0: -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\sum_{i=1}^n y_i - \sum_{i=1}^n \beta_0 - \sum_{i=1}^n \beta_1 x_i = 0$$

$$\bar{y} = \frac{1}{n} \sum y_i$$

$$n\bar{y} - n\beta_0 - n\beta_1\bar{x} = 0$$

$$\bar{y} - \beta_1\bar{x} = \beta_0$$

$$\beta_1: \sum_{i=1}^n y_i x_i - \beta_0 \sum_{i=1}^n x_i - \beta_1 \sum_{i=1}^n x_i^2 = 0$$

$$\sum_{i=1}^n y_i x_i - (\bar{y} - \beta_1\bar{x}) \sum_{i=1}^n x_i - \beta_1 \sum_{i=1}^n x_i^2 = 0$$

$$\sum_{i=1}^n x_i y_i - n\bar{y}\bar{x} + n\beta_1\bar{x}^2 - \beta_1 \sum_{i=1}^n x_i^2 = 0$$

$$\sum_{i=1}^n x_i y_i - n\bar{y}\bar{x} = \beta_1 \sum_{i=1}^n x_i^2 - n\beta_1\bar{x}^2$$

$$\sum_{i=1}^n x_i y_i - n\bar{y}\bar{x} = \beta_1 \left( \sum_{i=1}^n x_i^2 - n\bar{x}^2 \right)$$

$$\frac{\sum_{i=1}^n x_i y_i - n\bar{y}\bar{x}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2} = \hat{\beta}_1 \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

# Ordinary Least Squares Regression